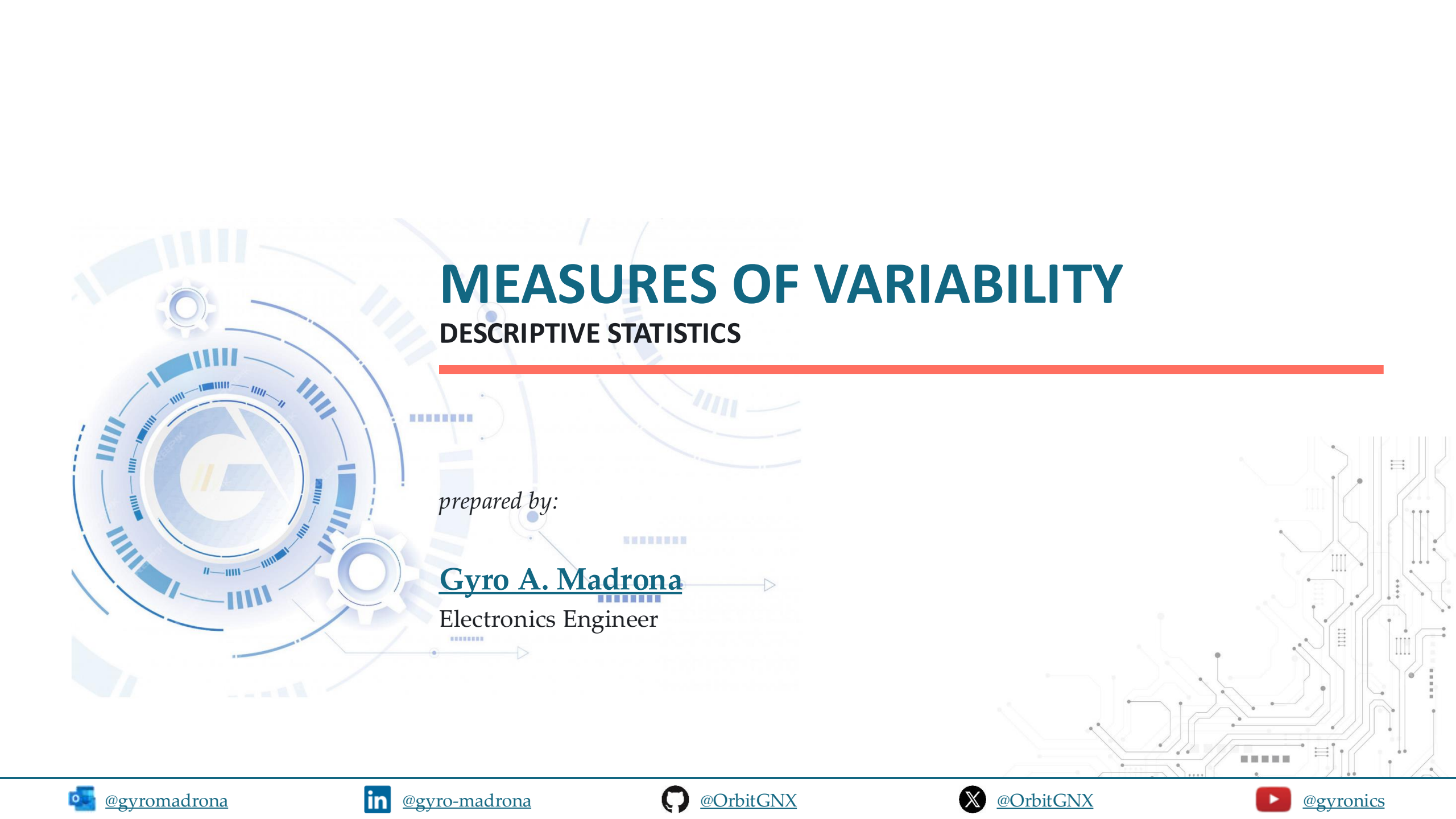




MEASURES OF VARIABILITY

DESCRIPTIVE STATISTICS

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TOPIC OUTLINE

Measures of Variability

Range and Interquartile Range

Variance and Standard Deviation

Coefficient of Variation



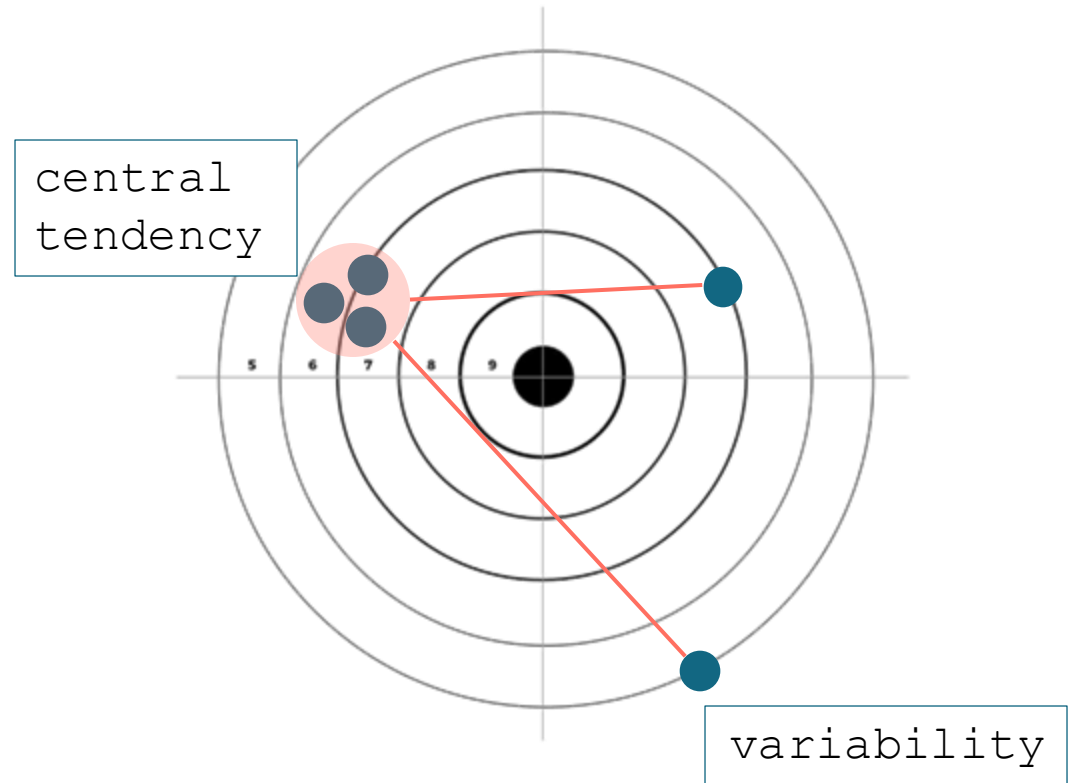
MEASURES OF VARIABILITY



MEASURES OF VARIABILITY

Measures of variability (or dispersion) describe how **spread out** or scattered the data points relative to the central tendency (mean, median, or mode).

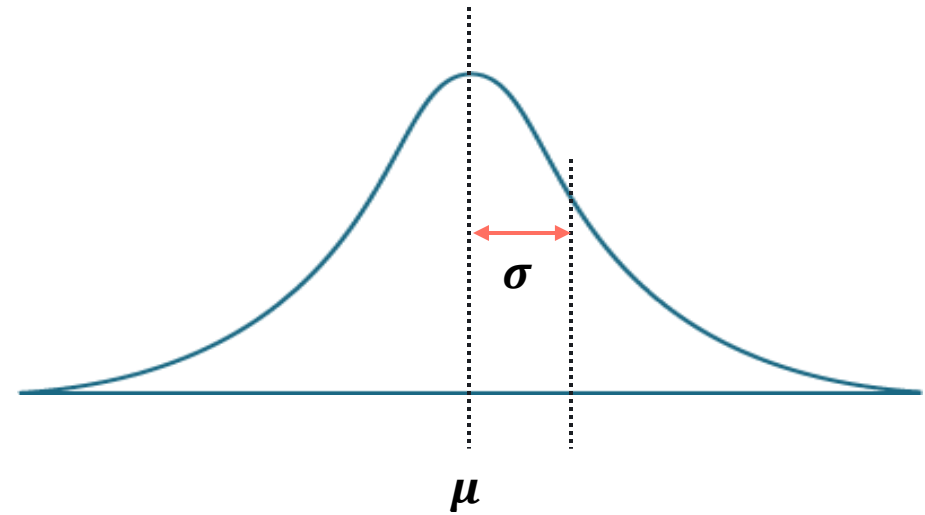
Dartboard Analogy



MEASURES OF VARIABILITY

Measures of variability (or dispersion) describe how **spread out** or scattered the data points relative to the central tendency (mean, median, or mode).

Normal Distribution



RANGE AND INTERQUARTILE RANGE



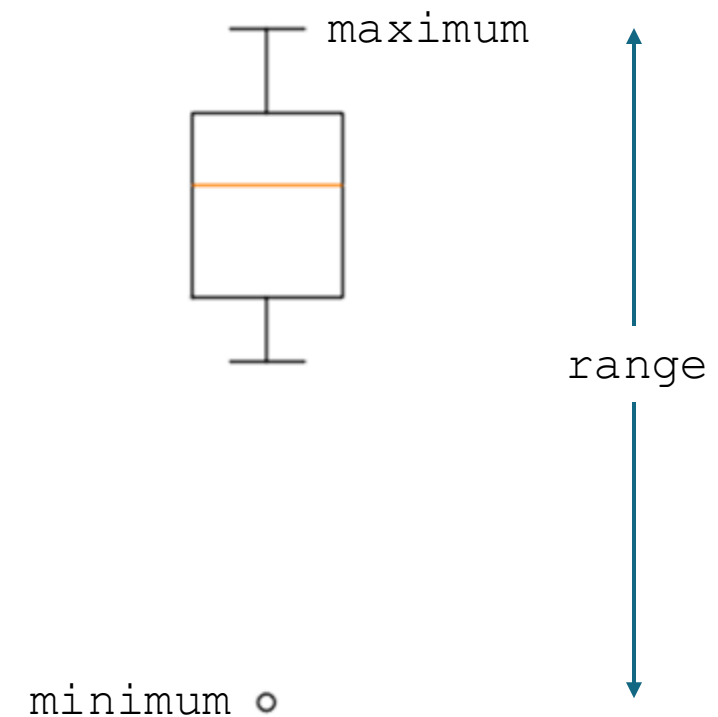
RANGE

Range is the simplest measure of variability and is calculated as the difference between the maximum and minimum values in a dataset.

Formula

$$\text{range} = \text{maximum value} - \text{minimum value}$$

Boxplot



INTERQUARTILE RANGE

The interquartile range (IQR) measures the spread of the middle 50% of the data, reducing the influence of outliers.

Formula

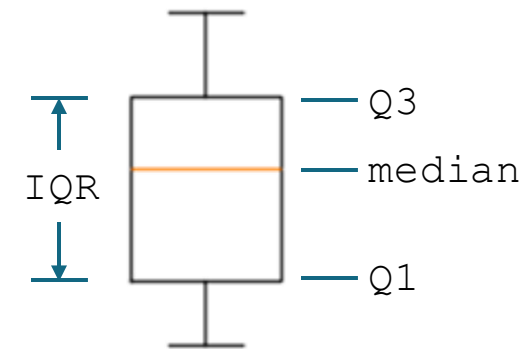
$$IQR = Q_3 - Q_1$$

where

Q_1 (first quartile) is the median of the lower half of the data (25%).

Q_3 (third quartile) is the median of the upper half of the data (75%).

Boxplot



OUTLIERS

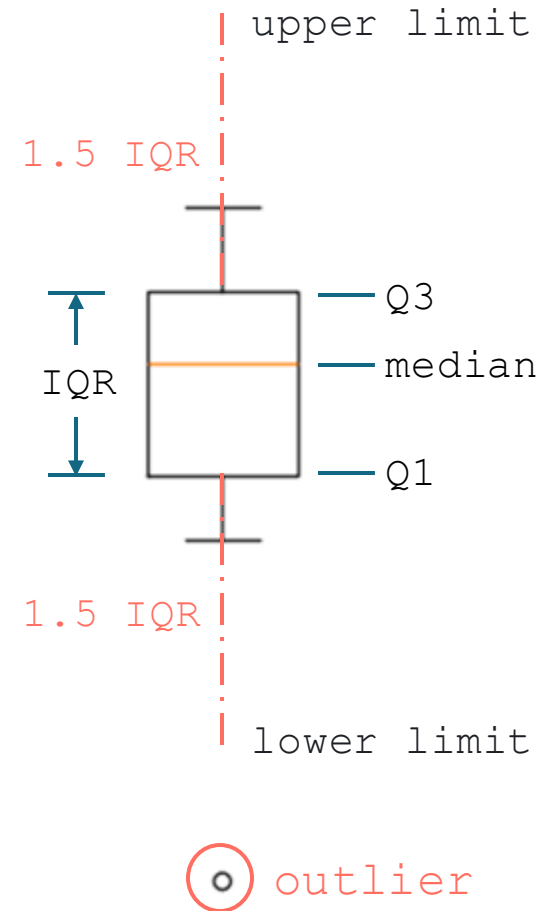
Outliers are data points that lie significantly outside the typical range of the rest of the dataset.

Formula

$$\text{Lower limit} = Q_1 - 1.5 IQR$$

$$\text{Upper limit} = Q_3 + 1.5 IQR$$

Boxplot



EXERCISE

The dataset provided contains the exam grades of 12 students. Calculate the range and interquartile range (IQR) to analyze the spread and variability of the grades.

solution

$$\text{Range} = \text{max} - \text{min}$$

$$\text{Range} = 9.8 - 3.5$$

$$\text{Range} = 6.3$$

ans

Exam Performance

Student	Grade
1	3.5
2	6.7
3	7
4	7.4
5	7.8
6	8.2
7	8.5
8	8.8
9	9
10	9.1
11	9.4
12	9.8



EXERCISE

The dataset provided contains the exam grades of 12 students. Calculate the range and interquartile range (IQR) to analyze the spread and variability of the grades.

solution

for Q_1

$$pos_{Q_1} = p(n-1) + 1$$

$$pos_{Q_1} = 0.25(12-1) + 1$$

$$\underline{pos_{Q_1} = 3.75}$$

$$Q_1 = x_i + f(x_{i+1} - x_i)$$

$$Q_1 = 7 + 0.75(7.4 - 7)$$

$$\underline{Q_1 = 7.3}$$

Exam Performance

Student	Grade
1	3.5
2	6.7
3	7 x_i
4	7.4 x_{i+1}
5	7.8
6	8.2
7	8.5
8	8.8
9	9
10	9.1
11	9.4
12	9.8



EXERCISE

The dataset provided contains the exam grades of 12 students. Calculate the range and interquartile range (IQR) to analyze the spread and variability of the grades.

solution

for Q_2

$$Pos_{Q_2} = P(n-1) + 1$$

$$Pos_{Q_2} = 0.5(12-1) + 1$$

$$Pos_{Q_2} = 6.5$$

$$Q_2 = X_i + f(X_{i+1} - X_i)$$

$$Q_2 = 8.2 + 0.5(8.5 - 8.2)$$

$$Q_2 = 8.35$$

Exam Performance

Student	Grade
1	3.5
2	6.7
3	7
4	7.4
5	7.8
6	8.2 X_i
7	8.5 X_{i+1}
8	8.8
9	9
10	9.1
11	9.4
12	9.8



EXERCISE

The dataset provided contains the exam grades of 12 students. Calculate the **range** and **interquartile range** (IQR) to analyze the spread and variability of the grades.

solution

$$pos_{Q_3} = P(n-1) + 1$$

$$Q_3 = X_i + f(X_{i+1} - X_i)$$

$$pos_{Q_3} = 0.75(12-1) + 1$$

$$Q_3 = 9 + 0.25(9.1 - 9)$$

$$\underline{pos_{Q_3} = 9.25}$$

$$\underline{Q_3 = 9.025}$$

Exam Performance

Student	Grade
1	3.5
2	6.7
3	7
4	7.4
5	7.8
6	8.2
7	8.5
8	8.8
9	9 X_i
10	9.1 X_{i+1}
11	9.4
12	9.8



EXERCISE

The dataset provided contains the exam grades of 12 students. Calculate the range and interquartile range (IQR) to analyze the spread and variability of the grades.

solution

$$IQR = Q_3 - Q_1$$

$$IQR = 9.025 - 7.3$$

$$IQR = 1.725$$

ans

$$\text{Lower limit} = Q_1 - 1.5 IQR$$

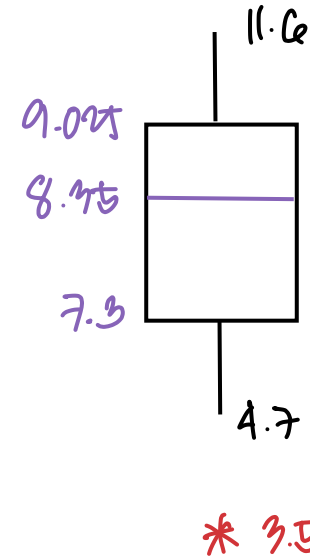
$$\text{Lower limit} = 7.3 - 1.5(1.725)$$

$$\text{Lower limit} = 4.7125$$

$$\text{Upper limit} = Q_3 + 1.5 IQR$$

$$\text{Upper limit} = 9.025 + 1.5(1.725)$$

$$\text{Upper limit} = 11.6125$$



Exam Performance

Student	Grade
1	3.5
2	6.7
3	7
4	7.4
5	7.8
6	8.2
7	8.5
8	8.8
9	9
10	9.1
11	9.4
12	9.8

Outlier



VARIANCE AND STANDARD DEVIATION



VARIANCE

Variance measures the average squared deviation of each data point from the mean.

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Sample Variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



STANDARD DEVIATION

Population Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

The standard deviation is the square root of variance.

Sample Standard Deviation

$$s = \sqrt{s^2}$$



EXERCISE

The dataset provided contains the sugar content (in grams) per serving for 10 popular breakfast cereals. Calculate the variance and standard deviation to measure the spread or variability in the sugar content across these cereals.

Breakfast Cereal	
Brand	Sugar
A	12
B	9
C	15
D	8
E	10
F	11
G	13
H	7
I	14
J	6

solution

Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{X} = \frac{12 + 9 + \dots + 6}{10}$$

$$\bar{X} = 10.5$$



EXERCISE

The dataset provided contains the sugar content (in grams) per serving for 10 popular breakfast cereals. Calculate the variance and standard deviation to measure the spread or variability in the sugar content across these cereals.

Breakfast Cereal		
Brand	Sugar	$(\bar{x} - x_i)^2$
A	12	2.25
B	9	2.25
C	15	20.25
D	8	6.25
E	10	0.25
F	11	0.25
G	13	6.25
H	7	12.25
I	14	12.25
J	6	20.25
		82.5

solution

Sample Variance

$$s^2 = \frac{\sum_{i=1}^n (\bar{x} - x_i)^2}{n-1}$$

$$s^2 = \frac{(10.5 - 12)^2 + (10.5 - 9)^2 + \dots + (10.5 - 6)^2}{10 - 1}$$

$$s^2 = \frac{82.5}{9}$$

$$s^2 = 9.17$$

ans

Sample Standard Deviation

$$s = \sqrt{s^2}$$

$$s = \sqrt{9.17}$$

$$s = 3.03$$

ans

POOLED STANDARD DEVIATION

Pooled standard deviation is a weighted **average** of the standard deviations from two or more groups.

Formula

$$\bar{\sigma}_{pooled} = \sqrt{\frac{\sigma_1^2 + \sigma_2^2 + \cdots \sigma_k^2}{k}}$$

where

k = number of groups

Variances add

$$\sigma_{total}^2 = \sigma_1^2 + \sigma_2^2 + \cdots \sigma_k^2$$

Standard deviations do not

$$\sigma_{total} \neq \sigma_1 + \sigma_2 + \cdots \sigma_k$$



EXERCISE

The dataset provided contains the battery life (in hours) for smartphones from different models.

Calculate the **pooled standard deviation** to measure the combined variability in battery life across these models.

solution

Average Variance

$$S_{ave}^2 = \frac{S_A^2 + S_B^2 + S_C^2}{k}$$

$$S_{ave}^2 = \frac{0.097 + 0.1 + 0.057}{3}$$

$$S_{ave}^2 = 0.08467$$

Pooled Stdev

$$S_{pooled} = \sqrt{S_{ave}^2}$$

$$S_{pooled} = \sqrt{0.08467}$$

$$S_{pooled} = 0.291$$

ans

Battery Life

Model	Hours
A	12.5
A	12.8
A	12.7
A	13.3
A	12.6
B	13.5
B	14.1
B	13.9
B	14.3
B	13.7
C	11.8
C	11.9
C	12.1
C	12.2
C	11.6

$$S_A^2 = 0.097$$

$$S_B^2 = 0.1$$

$$S_C^2 = 0.057$$



COEFFICIENT OF VARIATION



COEFFICIENT OF VARIATION

Population Coefficient of Variation

$$c_v = \frac{\sigma}{\mu}$$

Coefficient of variation (c_v) is a relative measure of variability, expressed as the **ratio** of the standard deviation to the mean.

Sample Coefficient of Variation

$$\hat{c}_v = \frac{s}{\bar{x}}$$



EXERCISE

The provided dataset includes ice cream prices listed in both USD and PHP. Calculate the **standard deviation** and **coefficient of variation** for each currency to analyze the variability in prices.

solution

for USD

$$\hat{C}_V(\text{USD}) = \frac{S_{\text{USD}}}{\bar{X}_{\text{USD}}}$$

$$\hat{C}_V(\text{USD}) = \frac{0.31}{3.95}$$

$$\hat{C}_V(\text{USD}) = 0.078$$

ans

for PHP

$$\hat{C}_V(\text{PHP}) = \frac{S_{\text{PHP}}}{\bar{X}_{\text{PHP}}}$$

$$\hat{C}_V(\text{PHP}) = \frac{17.80}{229.39}$$

$$\hat{C}_V(\text{PHP}) = 0.078$$

ans

Ice Cream Price List

Brand	Price (USD)	Price (PHP)
Brand A	3.5	203
Brand B	4	232
Brand C	3.75	217.5
Brand D	4.25	246.5
Brand E	3.9	226.2
Brand F	4.1	237.8
Brand G	3.6	208.8
Brand H	4.5	261
Brand I	3.8	220.4
Brand J	4.15	240.7

$$\bar{X}_{\text{USD}} = 3.95 \quad \bar{X}_{\text{PHP}} = 229.39$$

$$S_{\text{USD}} = 0.31 \quad S_{\text{PHP}} = 17.80$$

EXERCISE

The given dataset consists of test results from two machines, the Jaguar and Panther models, which produce **10 Ω** resistors with $\pm 5\%$ tolerance.

Determine which machine performs better based on its measures of variability for resistance values.

Resistance Test		
Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
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3	9.3	9.1
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5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
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EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
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7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
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3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
4	9.8	20
5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



EXERCISE

solution

Resistance Test

Test No.	Jaguar	Panther
1	10.6	10.1
2	9.1	11
3	9.3	9.1
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5	10.5	9.2
6	10.4	10.8
7	9.5	9.9
8	11	9.2
9	10.4	9.1
10	3	9.1
11	9.8	



LABORATORY

